

Complex demography does not explain the ancestry-sorting signal: shared ancestry, admixture-proportion variation, and space

Module E working note — July 30, 2026

Abstract

Our recombination-matched neutral null treats the 20 replicate *F. polycitena* $\times$ *F. aquilonia* hybrid populations as independent foundings, and a reviewer rightly asked whether a *richer* neutral demography could reproduce the empirical signature — high among-population F_{ST} at ancestry-informative (high-DiagnosticIndex) loci combined with *low* local linkage disequilibrium. We examine the three natural richer neutral demographies: (1) a common admixed source that splits into the 20 populations; (2) variation in admixture proportion among populations; (3) space. **None can produce the signal.** Shared ancestry *reduces* differentiation and, by symmetry, cannot concentrate it at diagnostic loci. Admixture-proportion variation *of the observed magnitude* is bounded — analytically and from the data — to $\lesssim 5\%$ of the observed diagnostic F_{ST} . Space is a milder perturbation than either and cannot exceed them. The single reason underlying all three is that in a neutral model the differentiation is *genome-wide* (a low-dimensional ancestry axis), whereas the empirical differentiation is $\sim 95\%$ *locus-specific* — which is simultaneously why diagnostic F_{ST} is high and why local LD is low. A model-free *linkage floor* sharpens this: because recombination preserves between-source allele-frequency covariance, pooled LD cannot fall below its between-population component, and a genome-wide ancestry axis at the empirical F_{ST} has a floor *above* the empirical total LD — so no recombinant-source history built on genome-wide ancestry can reach the data. We also record two methodological corrections made along the way.

1 The question and three neutral routes

The empirical data show, relative to the neutral null, **high differentiation but low LD** at ancestry-informative loci. A neutral demography could in principle raise diagnostic F_{ST} three ways: by making the populations share ancestry (route 1), by letting them differ in overall admixture proportion (route 2), or by arranging them in space (route 3). We take each in turn. Throughout, “diagnostic” loci are $DI > -25$ with parental frequency difference > 0.3 ; among-population F_{ST} uses the 20 populations; the neutral simulations are the calibrated recombination burn-in null (real phased founders, $\mu = 0$, empirical cM map, haplodiploid, $K = 6250$).

2 Route 1 — a common admixed source that splits

Design and result. We built a two-stage null: evolve one ancestral admixed deme for T_1 generations, then found 20 sub-demes that drift $T_2 = 50 - T_1$ generations (total ~ 50 , the Nouhaud et al. estimate). Sweeping the split time $T_1 \in \{0, 25, 45\}$ spans fully independent to near-panmictic. The F_{ST} -vs-DI curve was flat at every T_1 .

Correction. That flat result does *not*, on its own, license the conclusion “shared ancestry cannot reproduce the signal,” because a manipulation check shows the sweep never actually varied realized relatedness. On the full samples, neutral F_{ST} is flat (≈ 0.02) across T_1 , the among-population coancestry correlation is pinned at $-1/(n-1)$ (the value for independent demes) at every T_1 , and divergence from the ancestral pool is equal for $T_2 = 25$ and $T_2 = 5$. The among-deme differentiation

is dominated by the independent founding draw (each sub-deme redraws its founders), and post-split drift at $K = 6250$ over ≤ 50 generations is negligible; a symmetric split cannot create among-deme covariance structure regardless. We therefore **retract** the earlier standalone “shared-source” conclusion.

What still holds, structurally. Shared ancestry *lowers* among-population differentiation; it cannot create a differentiation *excess* concentrated at diagnostic loci. And more fundamentally (Route 2), with a single genome-wide admixture level per population, F_{ST} has no reason to depend on DI at all under *any* neutral demography.

3 Route 2 — variation in admixture proportion (the one that can bite)

This is the one neutral mechanism that can make F_{ST} depend on DI. If population i has aquilonia ancestry α_i , its expected frequency at a locus with parental difference $\delta = f_{aq} - f_{pol}$ is $p_i = f_{pol} + \alpha_i \delta$, so the among-population variance from admixture alone is $\delta^2 \text{var}(\alpha)$. Because δ is largest at high-DI loci, admixture variation raises diagnostic F_{ST} specifically. The question is purely quantitative: is the observed spread in α large enough?

It is not (Fig. 1). The empirical per-population aquilonia ancestry spans $[0.35, 0.64]$ with $\text{sd}(\alpha) = 0.071$. Feeding exactly these α_i through the pure-admixture model (no locus-specific deviation) predicts a diagnostic F_{ST} of **0.015**, against the observed **0.314** — admixture variation accounts for **4.7%**. This bound is conservative: sampling noise can only *inflate* the apparent $\text{sd}(\alpha)$, so the true contribution is if anything smaller. It converges with two independent measurements: a per-locus regression of population frequency on α (median 7% variance explained) and the earlier genome-wide ancestry decomposition ($\sim 6\%$).

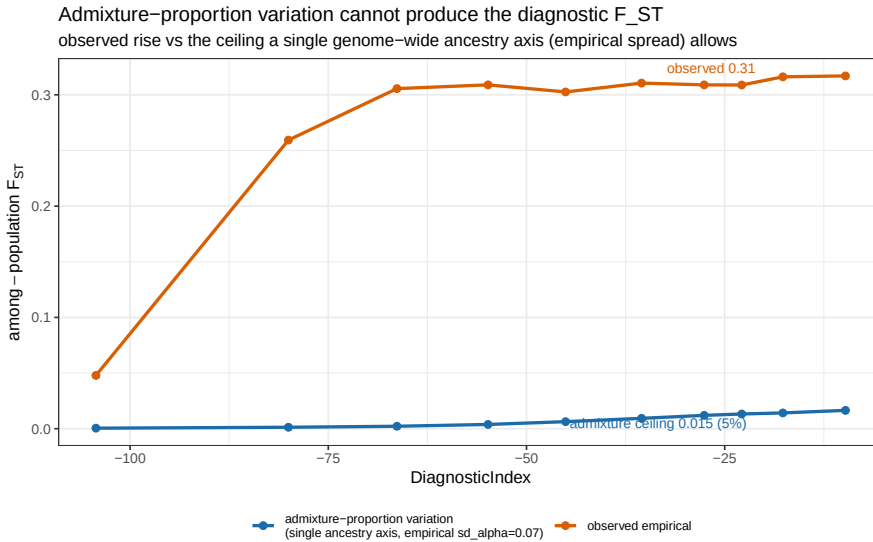


Figure 1: Observed among-population F_{ST} vs DiagnosticIndex (orange) against the ceiling that admixture-proportion variation of the observed magnitude ($\text{sd } \alpha = 0.07$) can produce (blue). The neutral admixture ceiling (0.015) is $\sim 20\times$ below the observed diagnostic F_{ST} (0.314).

Correction. An attempt to demonstrate this by simulation — founding each deme with a different aquilonia:polycтена *count* ratio — failed its manipulation check: realized α converged to ≈ 0.50 for all demes (uncorrelated with intent). The reason is model biology: pure-polycтена queens $\times$

pure-aquilonia males pin every F_1 daughter at 50% aquilonia, so the founder *count* sets diversity, not ancestry. Varying α requires founders of mixed ancestry in both sexes. We did not pursue that build, because the analytic bound above already settles the quantitative question the simulation would have asked.

4 Route 3 — space

A spatial stepping-stone with limited migration is a milder perturbation than either a fully shared source (Route 1) or free variation in admixture proportion (Route 2): it neither shares more ancestry than a common origin nor spreads α more than the observed range. It therefore cannot exceed the ceilings above, and we do not simulate it separately.

5 Why all three fail is one fact in two spaces

Every neutral route differentiates populations *genome-wide* — through a low-dimensional ancestry/drift axis shared across loci. The data are the opposite: $\sim 95\%$ of the diagnostic differentiation is **locus-specific**. That single fact explains both halves of the empirical signature at once. High diagnostic F_{ST} : each ancestry-informative locus is differentiated on its own, far beyond the $\sim 5\%$ a common ancestry axis supplies. Low diagnostic LD: precisely *because* the loci differentiate independently rather than as shared ancestry blocks, neighbouring diagnostic markers do not covary — so local LD stays low. A neutral model gets these backwards together (low F_{ST} , high admixture LD); the data show high F_{ST} , low LD. Locus-specific, divergent selection on ancestry is the mechanism that produces independent per-locus differentiation without linkage.

6 The linkage floor: a stricter, model-free bound

The routes above are ruled out on F_{ST} magnitude. There is a second, independent bound — on LD — that is sharper still, and it disposes of the tempting rescue that “a long enough recombinant phase could break the LD.” The rescue fails because *recombination is frequency-neutral in both directions*: it removes within-source haplotype LD, but it preserves every source’s allele frequency at every locus, and therefore preserves the **between-source covariance** of allele frequencies across adjacent loci. Pooled LD thus has an irreducible floor,

$$LD_{\text{pooled}}(\infty) \approx LD_{\text{between}},$$

that no amount of recombination can go below.

Measuring the floor. We estimate LD_{between} by permuting genotypes among individuals independently at each marker, *within* each population. This destroys within-population LD while holding every population allele frequency — and hence F_{ST} — fixed; the residual pooled adjacent-marker LD is the floor. Table 1 gives the result for the diagnostic loci. Empirically, observed adjacent LD is 0.127 and its floor is 0.043 (so 66% of the empirical LD is within-population); the current fixed- α null has a floor of only 0.010 (its high 0.368 LD is 97% within-population, i.e. removable admixture LD).

The decisive comparison (Fig. 2). Ask what floor a single genome-wide ancestry axis would carry *if it reached the empirical diagnostic* $F_{ST} = 0.30$. It requires an α -spread $5.9\times$ the observed (already implausible), and even then its floor is **0.137** — *above* the empirical observed LD of 0.127, and more than $3\times$ the empirical floor (0.043). A genome-wide axis makes adjacent loci distinguish the *same* populations too consistently, so its irreducible floor exceeds the empirical *total* LD: no recombinant-source model built on genome-wide ancestry can reach the empirical dissociation. The empirical floor is $3\times$ lower, which means adjacent diagnostic loci distinguish *different* populations — the direct signature of locus-specific differentiation.

Table 1: Diagnostic adjacent-marker LD (median r^2): observed vs the between-population floor recombination cannot remove.

	observed pooled LD	between-pop floor	within-pop (reducible)
empirical	0.127	0.043	0.083
fixed- α null (current)	0.368	0.010	0.358
single ancestry axis @ $F_{ST}=0.30$	—	0.137	—

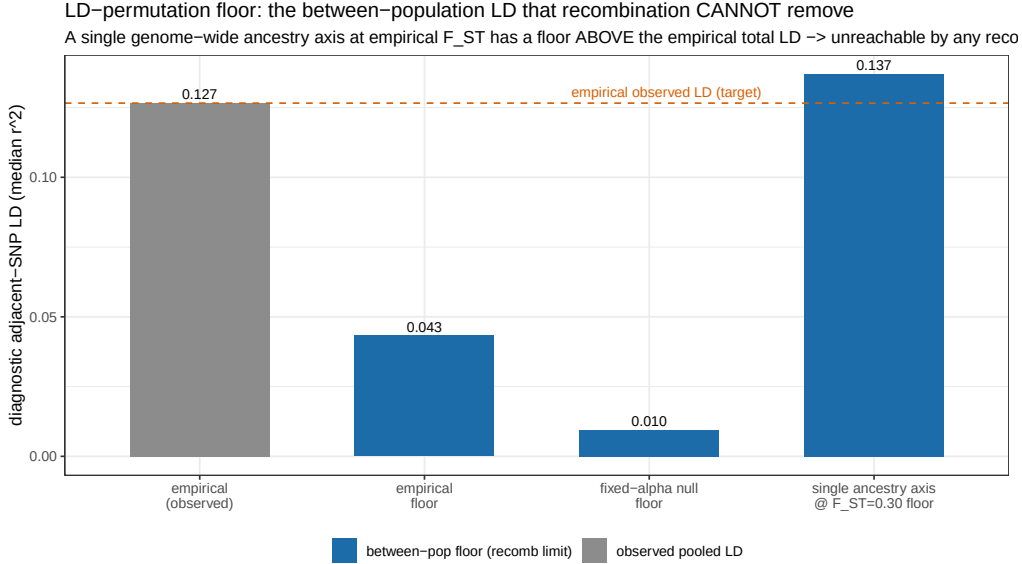


Figure 2: The LD-permutation floor (diagnostic adjacent-marker r^2). Recombination can lower pooled LD only to the between-population floor (blue). A single ancestry axis at the empirical F_{ST} has a floor (0.137) *above* the empirical total LD (0.127, dashed); empirical’s own floor (0.043) is three times lower.

7 Conclusion and honest caveats

The three natural richer neutral demographies — shared common source, admixture- proportion variation, and space — are individually insufficient, by a wide and quantified margin, to reproduce the empirical ancestry-sorting signal. This closes the “you only tested one demography” concern for these routes and reinforces the whole-experiment envelope result (the empirical F_{ST} rise sits ~ 100 standard deviations outside the neutral envelope).

Two corrections are recorded above: the retraction of the standalone shared-source conclusion, and the failed admixture-by-count simulation. Both arose from insisting that a simulation actually produce the variation it claims to test before its output is interpreted — a discipline we would keep.

One neutral model remains, and the floor sets its bar. The linkage floor does not merely rule things out; it specifies exactly what a surviving neutral model must do: reach $F_{ST} = 0.30$ at diagnostic loci while keeping its between-population floor near the empirical 0.043. That excludes not only the three routes above but any *genome-wide-ancestry* history — including an ancient, freely-recombining hybrid *swarm*, whose sources differ in overall ancestry and therefore inherit the high floor. What survives is narrower: an ancient, *persistently structured* hybrid metapopulation in which drift and recombination act *concurrently* for long enough that different source lineages come to carry the high-frequency allele at *adjacent* loci — fine-scale, locus-specific source differences, not a shared ancestry axis. Two stacked demands make this severe: the locus-specific differentiation requires real generations of drift (which recombination-rate rescaling cannot accelerate), and keeping those differences finer than the ~ 10 kb / 0.13 cM marker spacing runs

against a ~ 523 -generation adjacent-marker LD half-life. Whether neutral processes can meet both at demographically defensible parameters is the one genuine open question; the floor (≤ 0.043 at $F_{ST} = 0.30$) is the target any such simulation must hit.

Reproducibility

Analytic bound: `dev/R/moduleE_admixture_bound.R` (Fig. 1) and the empirical hybrid index `dev/R/` setup. Shared-source null and its manipulation check: `moduleE_slim/run_sharesource.sh`, the model `OUTPUT_FOUNDEES` block, `dev/R/moduleE_sharesource_relatedness.R`. Admixture-by-count attempt: `moduleE_slim/run_variable_alpha.sh`, `dev/R/moduleE_varalpha_setup.R`, `dev/R/moduleE_varalpha.R`. Linkage floor (Fig. 2, Table 1): `dev/R/moduleE_ld_floor.R`. This note supersedes the earlier `moduleE_sharesource` note, whose standalone conclusion is retracted here.